

The Generation-4 Single-Cell Collagen Model

Theory, equations and staged validation from elastic calibration to clutch failure and reaction-driven movement

G4 單細胞膠原模型：從彈性校準、間接傳力、clutch failure 到細胞移動

Why G4 exists	What G4 establishes
Earlier demonstrations mixed geometry, anchoring and contact choices, so visible differences could not be assigned cleanly to one cause. G4 fixes a single-cell elastic baseline first, then adds one mechanism per stage.	A connected bead-spring collagen network with probabilistic permanent crosslinks; local Gaussian contact; graph-traced indirect motion; stochastic Bell clutch slippage; and an unbiased reaction-driven moving-cell comparison.

KEY EQUATION

$$\zeta_b \, dr_i/dt = F_i^{\text{stretch}} + F_i^{\text{bend}} + F_i^{\text{xlink}} + F_i^{\text{repulsion}} + F_i^{\text{active}}$$

The common G4 backbone: inertia-free collagen beads move according to elastic, contact and active forces divided by bead drag.

Scope statement

G4 is a 2-D, one-cell mechanism-calibration platform in μm , nN and s. It is not a complete tumour-migration model. G4 v1 is preserved as the short-time prototype; G4 v2 extends observation time and makes small mechanical and stochastic events visible without inflating the physics.

One-sentence causal chain

Local cell pull → collagen stretch/bending → crosslink-mediated transmission → clutch loading and force-dependent rupture → traction imbalance → rigid-cell reaction motion.

Code: generations/g4_v2_multiscale/model.py · Theory generated 2026-09-07

1 · Positioning: G4 is a controlled calibration generation

Pre-G4 issue	G4 decision	What becomes interpretable
Few fibres or over-constrained ends	Use the G2-scale 99-fibre network with outer-boundary anchoring	Local deformation and force transmission can occur without whole-network drift
Every apparent crossing treated alike	Retain intersection links probabilistically	Link density/topology becomes an explicit experimental variable
Gaussian force could look long-range	Require a finite surface-contact shell before Gaussian weighting	Only physically local fibres receive direct force
Fixed and moving models differed in many ways	G4D releases only rigid-body translation/rotation	Motion is attributable to reaction-force imbalance
Individual clutch slips were invisible	Long overview + event-aligned 20 s microscope	Rupture, site failure, traction drop and recoil can be synchronized

Stage sequence

Stage	Cell	New test	Primary output
G4A	Fixed	One-factor elastic calibration	Displacement, traction, radial order and boundary control
G4B	Fixed	Crosslink-path transmission	Direct / one-hop / two-plus / unconnected motion
G4C	Fixed	Independent vs shared-load clutch failure	Bound count, traction, rupture cascade and recoil
G4D	Released	Reaction-driven translation/rotation	Path, displacement, traction imbalance and matrix response

G4 v1 versus G4 v2

- G4 v1: 2–5 minute interactive prototype that exposed the basic controls but made small one-hop displacements and fast rupture events difficult to see.
- G4 v2: the same single-cell concept and parameter language, but 2 h elastic/moving runs, a 30 min clutch overview, a 20 s event microscope, crosslink-graph statistics and explicit independent/shared-load comparison.

Clean inference rule

A stage may inherit everything already accepted, but its comparison should change one mechanism only. G4 does not use plasticity, proteolysis or nonlinear stiffening to rescue an unvalidated elastic baseline.

2 · Collagen mechanics shared by G4A–D

A collagen fibre is a chain of visible beads connected by axial bonds. Consecutive bond triplets resist bending. Crosslinks couple material points on different fibres. Only beads in the outer boundary band are anchored in the main condition.

EQUATION

$$r_i(t+\Delta t) = r_i(t) + (\Delta t / \zeta_b) F_i^{\text{total}}$$

Forward-Euler update for non-fixed beads; fixed outer-boundary beads remain at their initial positions.

Axial force with compression softening

EQUATION

$$e_{ij} = \ell_{ij} - \ell_{ij}^0; \quad F_{i \leftarrow j}^s = k e_{ij} e_{ij}; \quad k = EA/\ell^0 \text{ for } e \geq 0, \quad k = pEA/\ell^0 \text{ for } e < 0$$

Tension uses the full axial stiffness; compression uses $p=0.10$ to represent easy microbuckling.

Discrete bending

EQUATION

$$\Delta \kappa_i = (r_{i-1} - 2r_i + r_{i+1}) - \kappa_i^0; \quad U_b = \frac{1}{2} (EI/\ell^3) \|\Delta \kappa_i\|^2$$

The reference curvature is the initial fibre shape. The bending multiplier changes EI without changing axial EA .

Permanent material-point crosslink

EQUATION

$$p_a = (1 - \alpha_a) r_i + \alpha_a r_j; \quad f_x = k_x [(p_b - p_a) - \ell_x^0]$$

Each link resists separation of the two material points but is freely hinged. In G4 it never ruptures or re-forms.

No SLS and no Brownian force

Overdamped drag dissipates motion and controls approach time; elastic bonds/crosslinks store energy. G4 does not include a standard-linear-solid element, thermal noise or permanent remodeling.

Tune basic mechanics with one-factor comparisons

Keep the cell fixed and compare the same collagen geometry while changing one coefficient at a time.

Coefficient	Equation location	What increasing it should do
Pull F_{pull}	Active contact force	Increase direct loading and deformation, until other stiffness/geometry limits dominate
Bending multiplier	EI term in U_b	Make fibres harder to rotate or straighten
Collagen modulus E	Axial EA and base EI	Reduce bond extension and usually reduce local displacement
Bead drag ζ_b	$\zeta_b \dot{r} = \Sigma F$	Slow the transient response; it should not change the final elastic equilibrium
Crosslink probability p_x	Link-retention rule	Add more load paths, but may also constrain local rotation
Crosslink stiffness k_x	$f_x = k_x \delta x$	Make each retained connection transmit separation forces more strongly
Gaussian width σ_c	Contact weights	Spread the same finite total pull more evenly over eligible contacts

KEY EQUATION

link j retained $\Leftrightarrow u_j < p_x$, with the same stored u_j for every comparison

Nested topology: raising p_x adds links without reshuffling the links that existed at lower p_x .

Left versus right panels

- Left is the fixed baseline/control.
- Right is the selected one-parameter variant. It uses the same initial network, cell position, candidate intersections, simulation duration and seed.
- The special mobile-boundary control releases outer anchors on the right. It tests whether an apparent response is local deformation or rigid translation of the whole ECM.

Time design

G4 v2 runs the elastic overview for 2 h with sparse geometry frames. This follows the observation scale of Riching et al.; it avoids creating visibility by arbitrarily increasing force. Solver time step remains 0.05 s, so frame sampling changes display size but not physics.

G4A determines

A plausible elastic response regime and the sensitivity of each coefficient. It does not yet prove the path by which non-contact fibres move, nor does it contain clutch detachment.

4 · Hybrid cell–collagen coupling: contact first, Gaussian second

For every candidate fibre, the closest point on a bead-to-bead segment is computed. A point must first fall inside a finite cell-surface shell. Gaussian weights are then normalized only among eligible contacts.

KEY EQUATION

$$d_m = ||p_m - r_c|| - R; \quad 0 \leq d_m \leq w_c; \quad w_m = \exp(-d_m^2 / \sigma_c^2) / \sum_{n \in C} \exp(-d_n^2 / \sigma_c^2)$$

The contact shell is 0–3 μm and baseline $\sigma_c = 1.5 \mu\text{m}$. Gaussian weighting is local; it is not evaluated over the entire ECM.

EQUATION

$$F_m^{\text{active}} = F_{\text{pull}} w_m (r_c - p_m) / ||r_c - p_m||; \quad (F_i, F_j) = ((1-\alpha)F_m, \alpha F_m)$$

Prescribed G4A/B pull is divided among eligible material points, then mapped to the two host beads using linear interpolation.

Why not pull the nearest bead only?

- The physical contact may lie between two discrete beads; using a material point reduces bead-resolution artifacts.
- The interpolation preserves the applied force and its first moment along the segment.
- The finite contact shell prevents distant collagen from receiving direct active force merely because the Gaussian has a nonzero mathematical tail.

Cell-surface repulsion

EQUATION

$$\delta_i = \max(0, R + c - ||r_i - r_c||); \quad F_i^{\text{rep}} = k_{\text{rep}} \delta_i (r_i - r_c) / ||r_i - r_c||$$

A one-sided penalty prevents fibre beads from penetrating the rigid circular cell. It is not an adhesion law.

Display warning

The thin green lines in the G4 visualization are bead-displacement vectors, sometimes magnified 10× or 50×. They are not collagen fibres, force links or crosslinks. True collagen geometry remains at 1×.

Trace indirect realignment through the crosslink graph

Keep the cell fixed; direct fibres receive active force, while every non-contact fibre receives zero Gaussian force.

KEY EQUATION

$$d_{\text{graph}}(f, C_{\text{direct}}) \in \{0, 1, 2+, \infty\}$$

0=direct contact; 1=one retained crosslink away; 2+=longer connected path; ∞ =unconnected.

Why an indirect fibre can move

- A direct fibre deforms under the cell pull.
- Its crosslink material point separates from the linked material point unless the neighbor moves; the crosslink therefore produces an equal-and-opposite restoring force.
- That force enters the neighbor's bead balance, while the neighbor's own stretch and bending distribute the deformation along its contour.
- This chain repeats over multiple links, with decreasing or heterogeneous response depending on topology, stiffness and anchoring.

Orientation and displacement metrics

EQUATION

$$\theta_f = \frac{1}{2} \text{atan2}(\text{mean} \sin 2\theta_e, \text{mean} \cos 2\theta_e); \quad \Delta\theta_f = \frac{1}{2} \text{atan2}[\sin 2(\theta_f - \theta_f^0), \cos 2(\theta_f - \theta_f^0)]$$

Nematic angles treat a fibre pointing at θ and $\theta + \pi$ as the same orientation. Each graph class reports mean and 90th-percentile $|\Delta\theta|$.

EQUATION

$$u_f = \text{mean}_{i \in f} \|r_i - r_i^0\|$$

Each graph class reports mean, 90th-percentile and maximum fibre displacement.

Causal verification

- No-crosslink negative control: use the same geometry and direct pull but remove links.
- One-hop displacement/orientation should exceed the unconnected control when links are present.
- A displayed indirect force path must contain actual retained crosslinks; no hidden Gaussian force is allowed on indirect fibres.

G4B determines

Whether the implemented permanent elastic links can transmit a local active pull beyond the directly contacted fibres. It does not yet model detachment, crosslink failure or irreversible remodeling.

Motor-clutch loading and stochastic slippage

Replace deterministic attachment with effective clutches that bind, load, rupture and rebind while the cell remains fixed.

Motor and extension loading

KEY EQUATION

$$v_a = v_0 \max(0, 1 - F_{\text{site}}/F_{\text{stall}}); \quad dx/dt = \max(0, v_a - v_{\text{sub}}); \quad f = k_c x$$

The motor slows under load. A fibre material point yielding inward with speed v_{sub} reduces relative clutch loading.

Bell slip-bond rupture

EQUATION

$$k_{\text{off}}(f) = k_{\text{off}}^0 \exp(f/F_b); \quad P_{\text{off}} = 1 - \exp[-k_{\text{off}}(f)\Delta t]; \quad P_{\text{on}} = 1 - \exp(-k_{\text{on}}\Delta t)$$

There is no deterministic 'too long' threshold. Force continuously changes the rupture probability.

Constant clutch force	Extension f/k_c	Median lifetime $\ln 2/k_{\text{off}}(f)$
0 nN	0 μm	38.5 s
1.5 nN = F_b	0.75 μm	14.2 s
3.0 nN = $2F_b$	1.50 μm	5.2 s

Independent-clutch baseline

EQUATION

$$F_{\text{site}} = \sum_{j \text{ bound}} k_c x_j$$

Each bound clutch has its own extension and rupture draw. One rupture removes only that clutch; the site remains attached as long as at least one clutch is bound.

Effective parameters

N/site	k_c	k_{on}	$k_{\text{off}0}$	F_b	v_0	$F_{\text{stall/site}}$
12	2 nN/ μm	0.055 s ⁻¹	0.018 s ⁻¹	1.5 nN	0.025 $\mu\text{m/s}$	8 nN

Interpretation boundary

These are effective clutch bundles, not 12 counted integrin molecules and not a single-molecule fit. A clutch may detach below the landmarks or survive above them.

7 · G4C shared-load cluster and event microscope

The shared-load condition is displayed beside the independent baseline. It is an explicit alternative closure, not a silent replacement.

KEY EQUATION

$$f_i = F_{\text{site}} / i; \quad r_i = i k_{\text{off}}^0 \exp[F_{\text{site}} / (i F_b)]; \quad g_i = (N - i) k_{\text{on}}$$

With i bound clutches, f_i is the load per survivor, r_i is the total rupture hazard and g_i the total rebinding hazard.

Cascade logic

- One clutch ruptures: i decreases.
- The site load is redistributed among fewer survivors, raising force per clutch and therefore the Bell off-rate.
- Additional ruptures may follow as a cascade, but rebinding can rescue the site.
- A complete attachment loss is the first passage to $i=0$. Individual ruptures are slips, not disappearance of the entire cell-collagen link.

Shared-site stiffness assumption

EQUATION

$$K_{\text{site}} = N k_c \text{ while } i > 0; \quad F_{\text{site}} = K_{\text{site}} x_{\text{site}}$$

The code holds the fully-bound bundle stiffness while at least one effective clutch remains, then divides the site load among survivors. This assumes external compliance preserves site load during redistribution.

How the visualization exposes failure

View	Duration / sampling	What it is for
Overview	30 min	Multiple loading cycles, population traction and bound-clutch count
Event microscope	20 s at 0.1 s/frame	Individual rupture marks, bound spokes, site failure, traction drop and fibre recoil
Ensemble	200 isolated-site trials	Lifetime distribution and selection of a median-like event rather than an extreme animation

Common random stream

Independent/shared and fixed/moving cases use counter-addressed random proposals. Identical site/time/event addresses receive the same random number until mechanics changes the event acceptance, enabling paired comparisons.

Release the rigid cell without prescribing direction

Keep G4C shared-load clutch mechanics and release only translation and rotation.

KEY EQUATION

$$\gamma_c dr_c/dt = -\sum_m F_m^{\text{cell} \rightarrow \text{ECM}}$$

The cell receives the equal-and-opposite reaction to the traction it applies to collagen. No +x force, preferred velocity or polarization_probability is supplied.

EQUATION

$$\gamma_\theta d\theta/dt = \sum_m (a_m - r_c) \times (-F_m); \quad \gamma_\theta = \chi \gamma_c R^2$$

Surface anchor arms convert asymmetric site reactions into torque. χ is the rotational-drag factor.

What creates direction?

- The initial collagen geometry is heterogeneous even though the rules are symmetric.
- Sites find different fibre orientations and stiffness paths, producing different loading and lifetimes.
- Stochastic rupture/rebinding changes the spatial traction distribution.
- The vector sum of these reactions can remain imbalanced long enough to displace and rotate the cell.

Left panel	Right panel	Controlled difference
Shared-load clutch; cell centre and angle fixed	Same network and random stream; cell translation/rotation released	Only rigid-body degrees of freedom
Anchored ECM main condition	Mobile-boundary ECM control	Local cell motion versus whole-network drift

Contact-site update

As the cell/network moves, material-point contacts are refreshed periodically. In the current G4 implementation an unbound sector may update its point on the same candidate fibre; bound sites retain their current material attachment until they release.

G4D determines

Whether asymmetric clutch traction on the heterogeneous collagen can generate reaction-driven rigid-cell motion. It does not model deformable protrusions, a nucleus, cell polarity signaling or full 3-D migration.

9 · How to read the G4 visualization

Visual element	Meaning	Not to be confused with
Small dots along coloured curves	Collagen beads connected by axial bonds	Independent particles popping in/out
Orange fibre	Direct-contact graph class	A separate protrusion grown from the cell
Blue / purple fibre	One-hop / two-plus-hop linked classes	Direct Gaussian force recipients
Gold diamond + short connector	Permanent elastic crosslink	Clutch or displacement arrow
Thin green line	Bead displacement vector, possibly magnified 10×/50×	Collagen, crosslink or force
Clutch spokes	Currently bound effective clutches at a surface site	Permanent fibre-cell weld
Red rupture mark	Clutch unbinding event	Fibre or crosslink breaking
Dashed original cell outline / trail	Initial cell position or centre path	An additional cell

KEY EQUATION

$$\Delta r_i = r_i(t) - r_i(0); \text{ displayed endpoint} = r_i(0) + M\Delta r_i$$

M=1, 10 or 50 changes only the green vector length. True bead and fibre positions are never multiplied.

Why some differences still look small

- One-hop mean displacement can be 10–3–10–2 μm, much smaller than a pixel in a 180 μm full-field view.
- Long observation time reveals slow motion but cannot create a large equilibrium deformation when the elastic force is genuinely small.
- Individual clutch events may occur between coarse overview frames; the event microscope is required to see the causal sequence.
- A single slip may be mechanically buffered by 11 parallel clutches; the entire link disappears only at zero bound clutches.

Presentation rule

Always state whether geometry or only displacement vectors are magnified. Use local zoom and numerical μm/nN readouts before changing force parameters for visibility.

10 · Parameter provenance and calibration boundaries

Parameter	Baseline / range	Status
Domain / fibres	180 μm / 99	Inherited G2-scale fixture; not image-derived density
Cell radius	10 μm	Single-cell fixture; replace with experiment-specific segmentation
Fibre length / diameter	20–80 μm / 0.30 μm	Within Lee et al. effective fibre scale; not molecular fibril dimensions
Collagen modulus	3 MPa	Softened mechanism-first choice; not tissue-matched fit
Bending multiplier	0.10, 0.25, 0.50, 1.0	One-axis sensitivity separated from axial stiffness
Crosslink probability	0, 0.20, 0.35, 0.60	Topology sensitivity; a 2-D crossing is not automatically a biological weld
Crosslink stiffness	5, 10, 25, 75 nN/ μm	Effective link compliance sensitivity
Bead drag	90, 180, 360 nN·s/ μm	Transient-response sensitivity; fit to time-resolved displacement
Total pull	12, 24, 48 nN	Mechanism range; not a fitted cell traction
Contact shell / σc	3 μm / 1.5 μm	Professor-approved hybrid starting choice
Cell drag	600 nN·s/ μm	Effective mobility; ensemble calibration required

What each paper supports

Source	Used for	Does not establish
Lee 2014	Discrete collagen fibres and effective size range	The exact G4 topology or density
Abhilash/Wang 2014	Alignment and long-range force transmission in connected fibre networks	That every 2-D crossing is crosslinked
Notbohm 2015	Compression-softening/microbuckling mechanism	The exact ratio $p=0.10$
Bell 1978	Exponential slip-bond off-rate	The G4 effective nN clutch parameters
Bangasser & Odde 2013	Motor-clutch load-and-fail coupling	A unique shared-load closure
Erdmann & Schwarz 2004; Gao 2011	Load sharing, rebinding and cluster failure	That G4's bundle stiffness assumption is molecularly exact
Riching 2014	2 h observation and 4 min imaging scale	The G4 force, drag or 2-D network

11 · Verification gates, limitations and references

Gate	Expected result
Time-step convergence	$\Delta t=0.05$ and 0.025 s preserve the qualitative result
No bound clutch	Traction equals zero
No crosslinks	Non-contact transmitted displacement collapses toward unconnected control
Graph transmission	One-hop response exceeds the unconnected class and has a traceable link path
Shared-load hazard	For fixed site load, fewer bound clutches increase per-clutch force and off-rate
Complete site failure	Cell-ECM site disappears only when bound count reaches zero
Event consistency	Rupture time matches event log, traction drop and animation frame
Fixed/moving pair	Same network, links and counter-random proposals; only cell degrees of freedom differ

Limitations

- 2-D projection; no 3-D pore geometry or out-of-plane fibre passage.
- One rigid circular cell; no deformation, nucleus, explicit protrusion shape or biochemical polarity field.
- Permanent linear crosslinks; no crosslink turnover, fibre sliding, damage or plastic memory.
- No collagen strain-stiffening, SLS relaxation, proteolysis, proliferation or Brownian forcing.
- Effective parameters are mechanism-first and require fitting to cell-specific traction, displacement and slip/recoil data.

Key references

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What G4 can claim

G4 is a staged, controlled single-cell platform that separates elastic parameter sensitivity, crosslink-mediated indirect motion, clutch failure and reaction-driven rigid-cell motion. It is not yet a predictive model of whole-tumour invasion.